Customer Segmentation and Churn Prediction for an Online Retailer

RFM Clustering, Classification with SMOTE, and Retention Strategy

Data Mining and Knowledge Discovery (505043)
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Final Academic Report

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Dataset: Online Retail II (Kaggle), December 2009 to December 2011

Executive Summary

This project studies the customers of a UK online gift retailer, using 1,067,371 transaction lines from December 2009 to December 2011. After cleaning, 5,878 identified customers and £17.4 million of revenue remain. The work has two parts. First, customers were grouped into four behavioral segments with RFM analysis and K-Means. Second, a classification model was trained to predict which customers will stop buying in the next 90 days.

Main findings

- Revenue is highly concentrated: the top 20% of customers generate 77% of revenue, and one segment (20% of customers) brings 74%.
- Churn can be predicted with a ROC-AUC of about 0.81 on unseen customers. The tuned Random Forest and XGBoost reach 0.815; Logistic Regression reaches 0.811. The models are close, so a simple, explainable model is almost as good as an ensemble.
- Recency is the strongest predictor of churn in every method used. Churn rises from 27% among customers who ordered within the last 49 days to 85% among those who last ordered more than 326 days earlier. Order frequency and the typical gap between orders come next.
- Customers with one order churn 80% of the time, against 30% for customers with five or more orders. Converting first-time buyers into repeat buyers is the largest retention opportunity by customer count.
- SMOTE did not improve ROC-AUC, because churn is only mildly imbalanced (57% churners). It changed the balance between precision and recall, and it is documented in an ablation.

Recommended actions

- Protect the 40 high-value customers flagged as medium or high risk with personal outreach before the autumn peak; they are few enough for individual attention.
- Run automated, low-cost win-back emails for high-risk standard-value customers, and stop spending on them if they do not respond.
- Build a welcome and second-purchase journey for new customers, since the first repeat order is where most churn is decided.
- Validate every campaign with an A/B test before scaling; this report measures prediction quality, not campaign effect.

1. Introduction and Business Understanding

Online retailers compete on repeat business. Acquiring a new customer usually costs more than retaining an existing one, and in gift and wholesale retail a small group of repeat buyers typically carries most of the revenue. A retailer therefore needs to know which customers are valuable, which are drifting away, and where a limited marketing budget should go.

The company in this study sells mostly gift-ware to private customers and wholesalers. Three business concepts organize the analysis:

- **Retention:** the share of customers who order again. Its opposite, churn, is the share who stop ordering.
- **Basket value:** the money spent per order. It separates large wholesale orders from small retail orders and shapes how much a customer is worth.
- **Revenue concentration:** how much of the revenue depends on a small group of customers, which determines how much effort each group deserves.

Research questions. (1) Which behavioral segments exist among the customers, and how much revenue does each bring? (2) Can the customers who will not return in the next 90 days be predicted from their purchase history? (3) Which behaviors drive churn, and what retention actions follow?

2. Data and Exploratory Analysis

2.1 Data description and quality

The Online Retail II dataset contains 1,067,371 invoice lines from 1 December 2009 to 9 December 2011, with 43 countries in the raw data. Each line has an invoice number, product code, description, quantity, invoice date, unit price, customer ID and country. Several quality problems had to be handled before analysis: 22.8% of lines have no customer ID (mostly guest checkouts), cancelled invoices (invoice numbers starting with C), non-positive quantities or prices, and duplicate lines. After removing them, 779,425 lines remain (73% of the raw data).

2.2 Key patterns

- **Seasonality.** Revenue climbs through autumn and peaks in November (highest month: November 2010, £1.17 million). September to November brings 38% of the revenue in the observed full months.
- **Geography.** The United Kingdom provides 83% of revenue; no other country is close.
- **Baskets.** Across 36,969 orders the median basket is £303 but the mean is £470, so a few very large orders pull the average up. Medians are therefore used throughout.
- **Customers.** 72% of customers ordered more than once, and the top 20% by spend generate 77% of revenue.
- **RFM relationships.** Frequency and Monetary are strongly related (Spearman correlation 0.86); Recency is negatively related to Frequency (-0.56), meaning recent buyers also tend to buy more often.

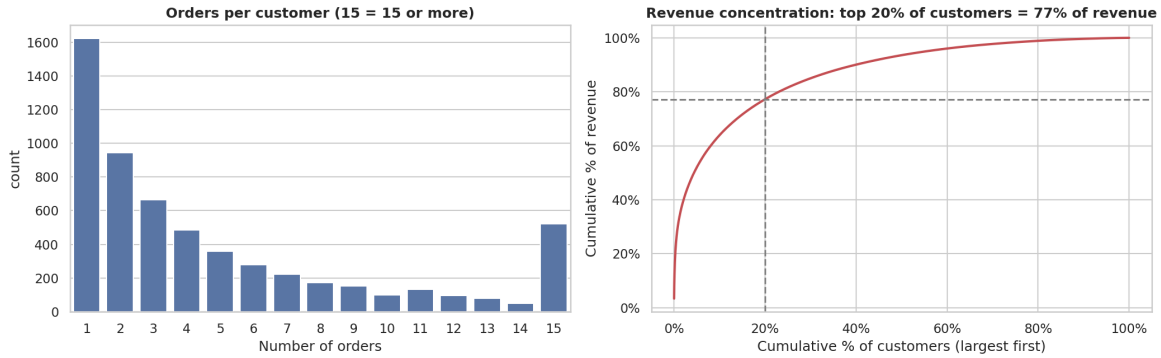


Figure 1. Orders per customer and revenue concentration (Pareto curve).

3. Data Preprocessing and Feature Engineering

3.1 Cleaning and RFM

Cleaning followed the rules in Section 2.1. For each customer, Recency is the number of days since the last order (measured from one day after the last invoice in the data), Frequency is the number of distinct invoices, and Monetary is total spend. These variables are strongly right-skewed (for example, the skewness of Monetary is 25.1), and K-Means uses Euclidean distance, so a log transformation (log1p) followed by standardization (StandardScaler) was applied. After the transformation the skewness of Monetary falls to 0.27. Large buyers are kept, not deleted, because they are the most valuable customers.

3.2 Defining churn without target leakage

A common shortcut defines churn as Recency above 90 days. If Recency is also an input feature, the model only has to rediscover that rule. We confirmed this: a Logistic Regression trained on that label with Recency as a feature reaches a test ROC-AUC of 1.000, which says nothing about real predictive ability.

Instead, a **forward-looking snapshot** was used. A cut-off date was set 90 days before the end of the data (10 September 2011). All features were computed only from transactions before the cut-off. A customer is labeled as churned (1) if they bought before the cut-off but made no purchase in the following 90 days. This yields 5,281 customers, of whom 2,989 (56.6%) churned. Recency measured at the cut-off is now a legitimate feature, because the label describes the future and the features describe the past.

3.3 Extended behavioral features

Feature	Description
Recency, Frequency, Monetary	Days since last order, number of orders, total spend before the cut-off
AvgBasketValue, AvgBasketItems	Average order value and units per order
UniqueProducts, ItemDiversity	Number of distinct products; distinct products per unit bought
Tenure	Days since the first purchase
AvgDaysBetweenOrders	Typical gap between orders (equal to tenure for one-order customers)
Orders_Last90d, Spend_Last90d	Activity in the 90 days before the cut-off
CancelRate	Share of the customer's invoices that were cancellations
IsUK	1 if the customer's main country is the United Kingdom

3.4 Train/test split and class imbalance

The data was split into 80% training (4,224 customers) and 20% test (1,057 customers), stratified by the churn label so both parts keep the 56.6% churn rate. The test set was not used for any decision except the final evaluation. SMOTE (Synthetic Minority Over-sampling Technique, Chawla et al., 2002) creates synthetic examples of the minority class by interpolating between neighbors. Here the minority class is the retained customers (43%). SMOTE was implemented as a step inside an *imblearn* pipeline, so during cross-validation it is fitted only on the training folds and never sees the validation fold or the test set.

4. Methodology

4.1 Customer segmentation with K-Means

K-Means was run on the scaled log-RFM matrix for K from 2 to 10, using k-means++ initialization with many restarts and a fixed random seed. K was chosen from the Elbow method (inertia), the Silhouette score, the Davies-Bouldin and Calinski-Harabasz indices, and business interpretability. The Silhouette score (Rousseeuw, 1987) is highest at K = 2 (0.439), but two clusters only separate active from lapsed customers. K = 4 is a local peak (0.365 against 0.347 at K = 3) and has the lowest Davies-Bouldin index among K of three or more (0.931). A stability check with ten random seeds gave a mean Adjusted Rand Index of 0.997.

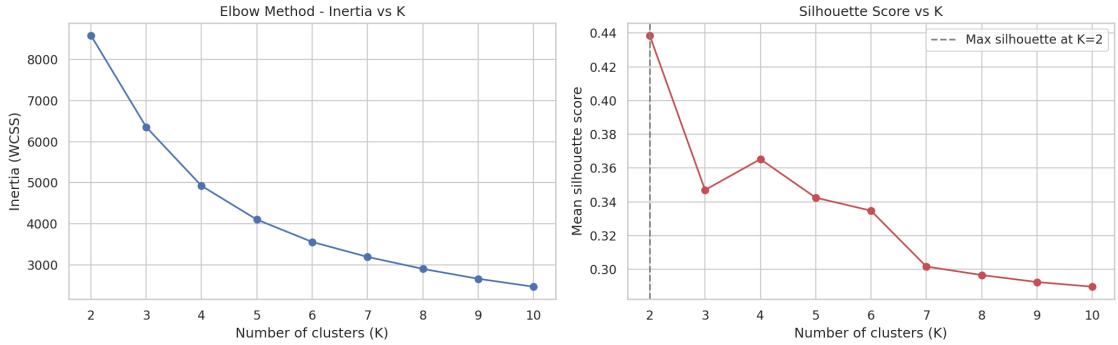


Figure 2. Elbow curve (left) and Silhouette score (right) for K = 2 to 10.

4.2 Churn classification

Four classifiers were compared: Logistic Regression, Decision Tree, Random Forest (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016), implemented with scikit-learn (Pedregosa et al., 2011). All four ran inside the same pipeline (log1p, StandardScaler, SMOTE, classifier). Hyperparameters were tuned with GridSearchCV using 5-fold stratified cross-validation on the training set and ROC-AUC as the scoring metric. The model with the best cross-validated ROC-AUC was selected without looking at the test set.

Model	Parameters searched	Best parameters
Logistic Regression	C: 0.01, 0.1, 1, 10	C = 10
Decision Tree	max depth 3, 5, 8, 12; min leaf 5, 20, 50; gini or entropy	entropy, depth 5, min leaf 20
Random Forest	trees 200, 400; max depth 8, 12, none; min leaf 3, 10	400 trees, depth 8, min leaf 10
XGBoost	trees 200, 400; depth 3, 5; learning rate 0.05, 0.1	200 trees, depth 3, rate 0.05

Evaluation used the confusion matrix, Precision, Recall, F1-Score, ROC-AUC and the precision-recall curve on the held-out test set. Feature importance was measured in five ways: Decision Tree and Random Forest impurity importance, XGBoost gain, Logistic Regression coefficients, and permutation importance on the test set.

5. Results

5.1 Customer segments

The final K-Means model (K = 4) has a Silhouette score of 0.365, a Davies-Bouldin index of 0.931 and a Calinski-Harabasz index of 5,058. The table below describes the four segments with median values per customer.

Segment	Customers	Revenue share	Last order	Orders	Spend
VIP / Champions	1,192 (20%)	74%	17 days	13	£4,995
Regular (cooling off)	1,465 (25%)	17%	185 days	5	£1,457
Recent / Potential	1,249 (21%)	6%	24 days	3	£727
At-Risk / Lost	1,972 (34%)	4%	403 days	1	£273

Table 3. Customer segments (medians per customer).

The second segment is named "cooling off" rather than "loyal" because its typical customer last ordered about six months ago.

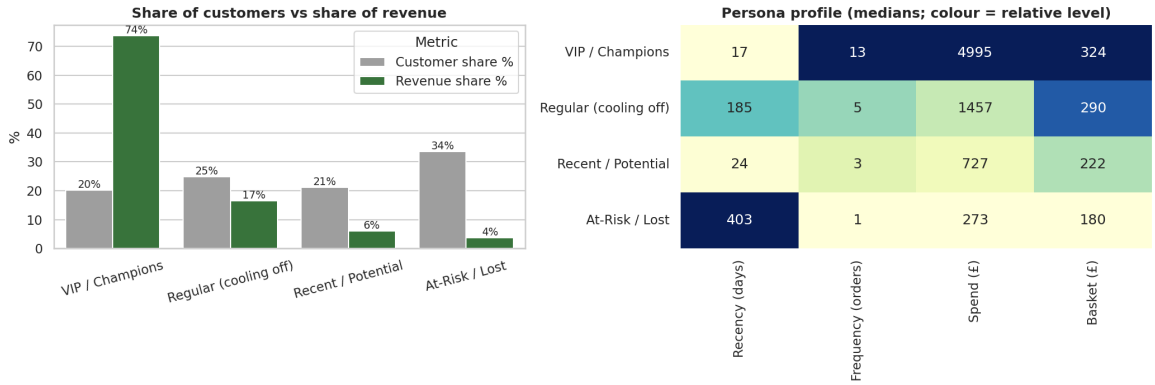


Figure 3. Segment shares and profiles.

5.2 Churn prediction performance

Model	CV ROC-AUC	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.797	0.755	0.789	0.774	0.781	0.815
XGBoost	0.793	0.751	0.794	0.756	0.775	0.815
Logistic Regression	0.791	0.745	0.797	0.736	0.765	0.811
Decision Tree	0.778	0.720	0.812	0.657	0.726	0.807

Table 4. Model comparison. CV ROC-AUC comes from training-set cross-validation; the other columns are from the test set (1,057 customers).

Three observations stand out. First, the four models are close: the test ROC-AUC ranges from 0.807 to 0.815. With 1,057 test customers, differences below 0.01 are within sampling noise, so no model is clearly better than the others. Second, tuning matters mostly for trees: the default Decision Tree reached a cross-validated ROC-AUC of 0.631 and the tuned tree 0.778, while Logistic Regression was already near its tuned score (0.791). Third, a majority-class baseline has a ROC-AUC of 0.5, so every model learns real signal. The Random Forest has the best cross-validated ROC-AUC and is the selected model; its test performance (0.815) is higher than its cross-validated score (0.797), so there is no sign of overfitting.

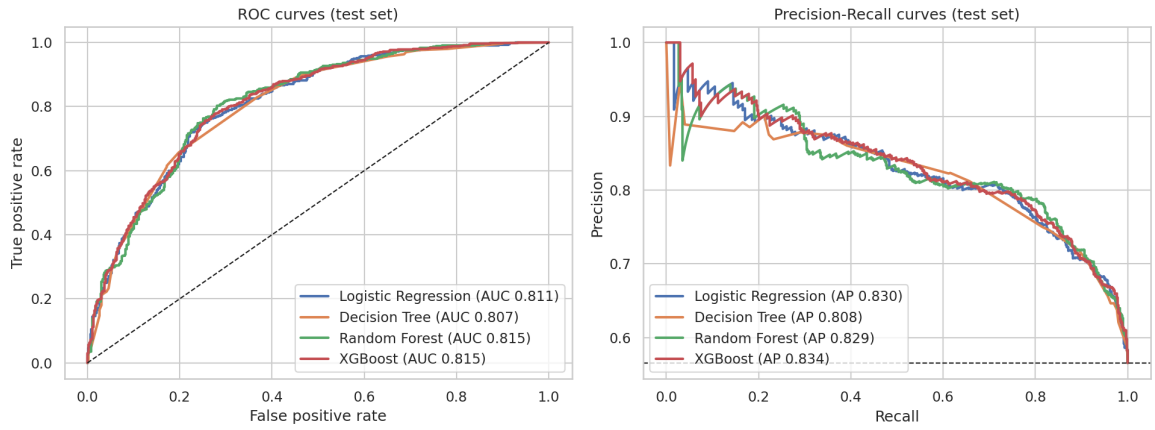


Figure 4. ROC and precision-recall curves on the test set.

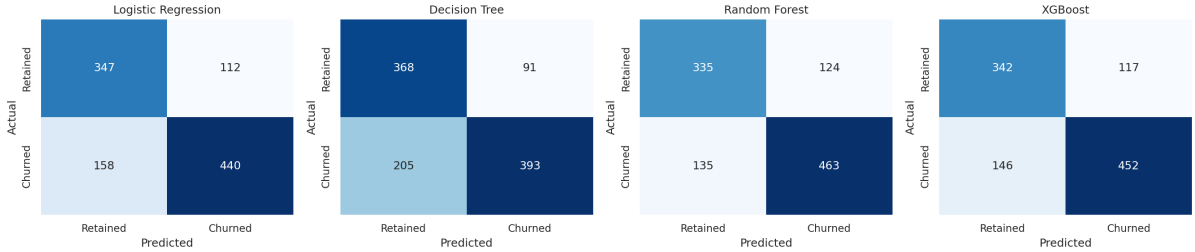


Figure 5. Confusion matrices at the 0.5 threshold (test set).

For the Random Forest at the default threshold, 463 of the 598 churners were identified (recall 0.774), and 124 of the 459 loyal customers were wrongly flagged. The Decision Tree is the most precise (0.812) but misses more churners (recall 0.657).

5.3 Effect of SMOTE

Model	SMOTE	Precision	Recall	F1-Score	ROC-AUC
Random Forest	with	0.789	0.774	0.781	0.815
Random Forest	without	0.760	0.848	0.802	0.817
XGBoost	with	0.794	0.756	0.775	0.815
XGBoost	without	0.748	0.833	0.788	0.815
Logistic Regression	with	0.797	0.736	0.765	0.811
Logistic Regression	without	0.754	0.816	0.784	0.811
Decision Tree	with	0.812	0.657	0.726	0.807
Decision Tree	without	0.772	0.811	0.791	0.806

Table 5. SMOTE ablation on the test set, using the tuned hyperparameters of each model.

SMOTE changes ROC-AUC by 0.002 or less. It increases precision and lowers recall, because the model sees more synthetic retained customers and becomes more cautious about predicting churn. Without SMOTE, F1-Score is slightly higher for three of the four models (for example 0.802 against 0.781 for the Random Forest). Churn is the majority class (57%), so oversampling is not necessary in this problem. It is kept in the pipeline because the project requires class-imbalance handling, and the ablation shows that its effect is a shift of the decision balance rather than a gain in ranking quality. Where missed churners are costlier than wasted offers, the pipeline without SMOTE (higher recall) is preferable.

5.4 Feature importance

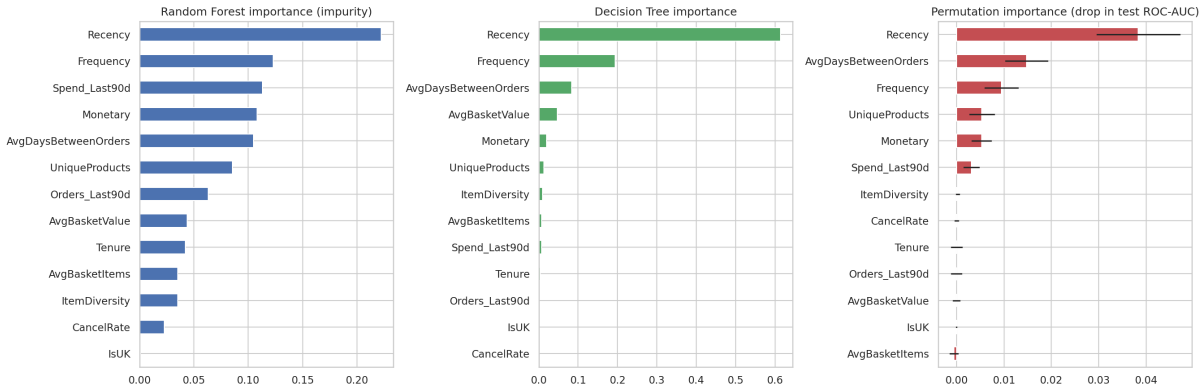


Figure 6. Feature importance from the Random Forest, the Decision Tree and permutation importance.

Recency is the dominant predictor in every view. It holds 61% of the Decision Tree importance, 22% of the Random Forest importance and 29% of the XGBoost gain. Shuffling Recency lowers the test ROC-AUC by 0.038, about 2.6 times the effect of the next feature (average days between orders, 0.015). Frequency, recent spend and total spend follow.

The direction of the effect is clear from the data:

- Churn rises with Recency: 27% for customers whose last order was within 49 days, 47% for 50 to 163 days, 68% for 164 to 326 days and 85% beyond 326 days.
- Churn falls with order count: 80% for one-order customers, 63% for two to four orders, 30% for five or more.
- Customers with no order in the 90 days before the cut-off churn 71% of the time, against 31% for those who ordered.
- Churn falls with total spend: 79% in the lowest spend quartile and 25% in the highest.
- IsUK, CancelRate, Tenure and basket size add almost nothing once the other features are present. Cancellation rate is also confounded with customer size, because heavy buyers cancel more often.

Logistic Regression coefficients are not interpreted one by one, because Monetary, number of products and basket value are correlated, which makes individual coefficients unstable (for example, Monetary -1.76 and AvgBasketValue +0.94). The group rates above and the permutation ranking are the more reliable evidence.

6. Business Strategy and Practical Application

6.1 Churn risk tiers

The selected Random Forest gives each test customer a churn probability. Customers were placed in three risk tiers (low below 0.4, medium 0.4 to 0.7, high 0.7 or above) and two value tiers (high value means total spend in the top quarter of the training customers). The table also shows the churn that actually happened, which validates the tiers.

Risk tier	Value	Customers	Share of past spend	Actual churn rate
Low (below 0.4)	High value	221	77.3%	15%
Low (below 0.4)	Standard	142	4.8%	36%
Medium (0.4-0.7)	High value	30	5.2%	43%
Medium (0.4-0.7)	Standard	360	8.6%	68%
High (at least 0.7)	High value	10	1.2%	90%
High (at least 0.7)	Standard	294	2.8%	85%

Table 6. Risk and value tiers on the test set (1,057 customers).

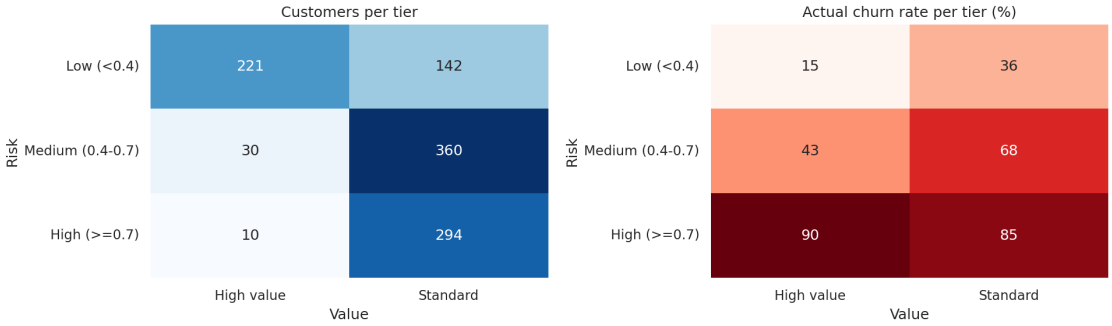


Figure 7. Customers per tier and actual churn rate per tier.

The tiers separate customers well: actual churn is 15% in the low-risk, high-value group and 90% in the high-risk, high-value group. Most high-value customers are safe: 221 of the 261 high-value customers have low risk, and together they hold 77% of the test group's past spend. So the personal retention effort concerns a small group: 10 high-risk and 30 medium-risk high-value customers. Most churn volume comes from standard-value customers, where cheap automated tactics are the right tool.

6.2 Recommended campaigns

Target group	Goal	Action	KPI
High-value, medium or high risk (40 customers in the test group)	Protect	Personal outreach or account call, tailored offer, check for service problems such as cancellations, early access to autumn stock	Retention rate of the group; revenue per customer
VIP / Champions segment	Reward	Loyalty tier, early access to new collections, thank-you contact	Order frequency
Regular (cooling off)	Win back	Reminder timed to the customer's usual reorder gap, restock alerts, modest incentive	Share who reorder within 60 days
Recent / Potential, one-order customers	Convert	Welcome series, second-purchase offer, recommendations based on the first basket	Second-order rate
High-risk, standard value	Low-cost reactivation	One or two automated win-back emails; stop paid spend if there is no response	Win-back response rate, cost per reactivated customer
Low risk, standard value	Do not discount	Standard lifecycle emails only; these customers are likely to order anyway	Margin protection

Timing matters: revenue peaks in autumn, so retention campaigns should run before September, and emails to business buyers are best sent on weekday mornings before the midday order peak (almost no orders are placed on Saturdays). The effect of these actions has not been measured. Each should be tested with a randomized A/B test (a treated group and a held-out control group within the same risk tier) before it is scaled.

7. Limitations and Future Work

- **Single cut-off.** The model uses one snapshot (10 September 2011). The 90-day outcome window includes the pre-Christmas peak, which lowers churn compared with a quiet season. Repeating the experiment with several cut-off dates and validating on a later period would test stability over time.
- **Random customer split.** Training and test customers come from the same period. A temporal validation would be a stricter test of real-world use.
- **Limited features.** Only purchase history is used. Product categories, marketing contacts, price promotions and customer type (wholesale or retail) are not available.
- **Identified customers only.** About 23% of lines have no customer ID and are excluded, so the results describe registered customers.
- **Persona and churn use different time frames.** The segments use the full period, whereas the churn model uses data before the cut-off, so a cross-tabulation of segment and churn would be circular and was not computed.
- **Prediction is not causation.** The model shows who is likely to churn, not whether a campaign will prevent it.
- **Model differences are small.** No model is significantly better than the others. A formal comparison (for example, a repeated-split paired test) was not performed.

8. Conclusion

RFM and K-Means reveal a strongly concentrated customer base in which about one fifth of customers bring three quarters of revenue. A leak-free churn definition and a pipeline with SMOTE inside cross-validation allow churn to be predicted with a ROC-AUC of about 0.81, and recency, frequency and purchase rhythm explain most of the risk. The practical value lies in the combination: risk scores say who is leaving, value tiers say who matters, and the campaign table says what to do. The next step is to test these actions in controlled experiments.

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Appendix A. Questions and Answers for the Oral Defense

Q1. Why not define churn as Recency above 90 days?

If Recency is also a feature, the label is a function of the feature. We tested it: ROC-AUC was 1.000, which is leakage, not skill. Our label looks 90 days into the future and uses only past data for features.

Q2. How exactly is churn defined?

A customer who bought before 10 September 2011 and made no purchase in the next 90 days (to 9 December 2011) is a churner. 56.6% of the 5,281 customers churned.

Q3. Churn is 57%. Why use SMOTE at all?

The project requires imbalance handling, and the retained class is the minority (43%). We implemented it correctly inside the pipeline and showed with an ablation that it does not change ROC-AUC; it only trades recall for precision. Reporting that honestly is part of the result.

Q4. How do you prevent leakage with SMOTE?

SMOTE is a pipeline step, so in each cross-validation fold it is fitted on the training part only. The validation fold and the test set are never oversampled or used to fit the scaler.

Q5. Why ROC-AUC for tuning?

It does not depend on a threshold and ranks customers by risk, which is how a retention team would use the scores. We also report precision, recall, F1 and the precision-recall curve.

Q6. All four models score about 0.81. Why?

The signal is mostly in a few behavioral variables (recency, frequency, rhythm), which even a linear model captures. The gaps between models are below 0.01 AUC, within sampling noise for 1,057 test customers.

Q7. Which model would you deploy?

The Random Forest, because it had the best cross-validated ROC-AUC. Logistic Regression is a strong, transparent benchmark, and in practice its simplicity could justify choosing it.

Q8. Why K = 4 when Silhouette is highest at K = 2?

K = 2 only splits active from lapsed customers. K = 4 is a local Silhouette peak, has the best Davies-Bouldin index for K of three or more, is stable across seeds (ARI 0.997), and gives four personas that need different actions.

Q9. Why a log transformation?

RFM variables are heavily right-skewed (Monetary skewness 25). Without it, a few very large customers dominate distance-based clustering. Large buyers are kept, not removed.

Q10. What does feature importance tell the business?

Recency matters most: churn goes from 27% to 85% across recency groups. Frequency and the gap between orders follow. Basket size, country and cancellation rate add little.

Q11. Is the model overfitting?

Test ROC-AUC (0.815) is not lower than cross-validated ROC-AUC (0.797), and the held-out test set was used only once for the final evaluation.

Q12. What about the 23% of rows without a customer ID?

They cannot be linked to a customer, so they are excluded from customer-level analysis. The results describe identified customers only.

Q13. How would you prove the retention actions work?

With a randomized A/B test inside each risk tier: treated and control groups, measuring retention rate and incremental revenue against campaign cost.

Q14. What is the biggest weakness of this study?

A single cut-off date and a random customer split. A multi-cut-off, time-based validation would show how stable the model is across seasons.

Appendix B. Reproducibility

Notebooks: *preprocessing.ipynb* (cleaning and RFM), *kmeans_rfm_segmentation.ipynb* (segmentation), *midterm_eda_personas_summary.ipynb* (exploratory analysis and personas), *churn_features.ipynb* (churn target, features, split, SMOTE demonstration) and *churn_modeling.ipynb* (training, tuning, evaluation, feature importance, risk tiers). Random seed 42 is used throughout. Python libraries: pandas, scikit-learn 1.8, imbalanced-learn, XGBoost, Matplotlib, Seaborn, Plotly, Streamlit.